

Project Planning Phase

Project Planning Template (Product Backlog, Sprint Planning, Stories, Story points)

Date	07 July 2026
Team ID	
Project Name	Visualization Tool for Electric Vehicle Charge and Range Analysis
Maximum Marks	5 Marks

Product Backlog, Sprint Schedule, and Estimation (4 Marks)

Use the below template to create product backlog and sprint schedule

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team Members
Sprint-1	Data Integration & Preprocessing	USN-1	As a data engineer, I can import EV charging, vehicle, and battery datasets into MySQL and connect them to Tableau so all data is available for analysis.	5	High	
Sprint-1		USN-2	As a data engineer, I can remove duplicate records and handle missing/NULL values so the dataset is clean and reliable.	3	High	
Sprint-1		USN-3	As a data engineer, I can split the CheapElectricCars column into three columns and rename Split 3 to Brands so brand data is usable.	5	High	
Sprint-1		USN-4	As a data engineer, I can change the Address field's data type to Geographic Role so map-based visualizations work correctly.	3	Medium	
Sprint-1		USN-5	As a developer, I can validate and correct data types	4	Medium	

			across text, numeric, and categorical fields so calculations are accurate.			
Sprint-2	Charging Pattern & Usage Dashboard	USN-6	As a city energy planner, I can view a heatmap of EV charging demand by time of day and duration so I can identify peak charging hours.	5	High	
Sprint-2		USN-7	As a city energy planner, I can filter charging stations by region and connector type so I can plan new infrastructure.	5	High	
Sprint-2		USN-8	As a city energy planner, I can view charging stations plotted on a map of India so I can see their geographic distribution.	4	Medium	
Sprint-2		USN-9	As a user, I can view a bar chart of charging stations grouped by region and type.	3	Medium	
Sprint-2		USN-10	As a user, I can export the charging dashboard as a PDF report.	3	Low	
Sprint-3	Battery Performance & Range Dashboard	USN-11	As a fleet manager, I can compare predicted vs. actual driving range for each vehicle so I can spot underperforming vehicles.	5	High	
Sprint-3		USN-12	As a fleet manager, I can view efficiency (Wh/Km) by brand so I can identify top-performing EV brands.	5	High	
Sprint-3		USN-13	As a fleet	4	Medium	

			manager, I can view top speed by brand so I can compare vehicle performance.			
Sprint-3		USN-14	As a fleet manager, I can filter vehicles by brand, body style, and powertrain type.	3	Medium	
Sprint-3		USN-15	As a user, I can view calculated fields such as BodyStyle_Count and Car_Brands_India for aggregated analysis.	3	Low	
Sprint-4	EV Model Comparison & Story	USN-16	As a consumer, I can compare price ranges of different EV models available in India so I can choose within my budget.	5	High	
Sprint-4		USN-17	As a consumer, I can view different EV car models side-by-side so I can compare specifications.	5	High	
Sprint-4		USN-18	As a consumer, I can filter models by brand, price, and region for a personalized comparison.	4	Medium	
Sprint-4		USN-19	As a user, I can view a guided Tableau Story that walks through charging stations, pricing, and brand comparisons.	3	Medium	
Sprint-4		USN-20	As a user, I can view the final dashboards published to Tableau Public/Server for sharing.	3	Low	

Project Tracker, Velocity & Burndown Chart: (4 Marks)

Sprint	Total Story Points	Duration	Sprint Start Date	Sprint End Date (Planned)	Story Points Completed (as on Planned End Date)	Sprint Release Date (Actual)
Sprint-1	20	6 Days	08 Jul 2026	13 Jul 2026		
Sprint-2	20	6 Days	14 Jul 2026	19 Jul 2026		
Sprint-3	20	6 Days	20 Jul 2026	25 Jul 2026		
Sprint-4	20	6 Days	26 Jul 2026	31 Jul 2026		

Velocity:

Imagine we have a 10-day sprint duration, and the velocity of the team is 20 (points per sprint). Let's calculate the team's average velocity (AV) per iteration unit (story points per day): $AV = \text{sprint duration} / \text{velocity} = 20 / 10 = 2$

Burndown Chart:

A burn down chart is a graphical representation of work left to do versus time. It is often used in agile software development methodologies such as Scrum. However, burn down charts can be applied to any project containing measurable progress over time.